SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2_AT2 — VISUALIZATION DEVELOPMENT EXERCISE

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO2 – Development (BL4)	**Assessment:**	Assessment Tool 2 – Visualization Development
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO2 Statement:	*Analyze and synthesize multiple data views into a coherent, annotated dashboard that communicates a clear narrative insight.(BL4)*		
Student Name:	K. Amrutha Sandhya	**Reg. No.:**	192524270
Section / Batch:	AIDS	**Date:**	30-07-2026

Code of Conduct

I K.Amrutha Sandhya (192524270) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Amrutha

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must include at least two coordinated charts sharing a common axis or theme.
- Include at least one annotation calling out a key data point in your dashboard.
- Write a one-sentence narrative takeaway summarizing the insight your dashboard reveals.

Annexure A – VISUALIZATION DEVELOPMENT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Visualization Development Exercise

CO2 · BT4: Analyze ·

This rubric is used to assess student responses to the CO2-AT2 Visualization Development Exercise, in which students build a small multi-chart dashboard from given data, add at least one annotation, and write a narrative takeaway.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of which chart types to combine, how they relate to each other, and how annotation and narrative should be used to communicate insight.	Good understanding of chart combination and annotation/narrative technique, with only minor gaps.	Basic understanding of dashboard development, but with some misconceptions about which charts to combine or how to annotate them.	Poor understanding of core development concepts; combines charts or annotations that do not suit the data or the story being told.
Explanation	**25%**	The annotation and narrative takeaway are precise, well-justified, and clearly tied to a specific, correctly identified data point or pattern.	The annotation and narrative are clear and reasonably well-justified, with adequate connection to the data.	The annotation or narrative is present but vague, generic, or weakly connected to the actual data shown.	Annotation or narrative is missing, incorrect, or unconnected to the data in the dashboard.
Application	**20%**	Effectively applies multi-chart coordination, annotation, and storytelling techniques to	Applies development techniques to produce a workable dashboard, with only minor errors or missed	Limited application; the dashboard partially addresses the scenario but charts feel disconnected or the story is unclear.	Fails to apply appropriate development techniques; the charts, annotation, and narrative do not

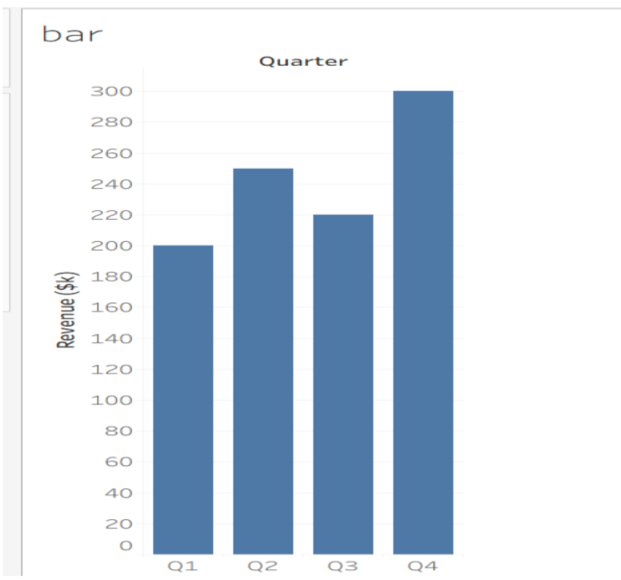
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		produce a complete, cohesive dashboard for the scenario.	refinements.		work together as a dashboard.
Organization	**15%**	Charts are clearly coordinated (aligned axes or shared theme), laid out logically, with the annotation and narrative placed for maximum clarity.	Charts are organized and mostly coordinated, with only minor issues in alignment or layout.	Basic structure; charts are present but poorly aligned or the annotation/narrative feels disconnected from the layout.	Poorly organized; charts are not coordinated, and the annotation or narrative is missing or misplaced.
Accuracy	**10%**	All chart values, scales, and the annotated data point are completely accurate and correctly reflect the given data.	Mostly accurate, with only minor omissions or a small error in one chart or the annotation.	Partially correct; some plotted values or the annotated point are missing or incorrect.	Largely inaccurate; major errors in plotted values, scales, or the annotated data point.

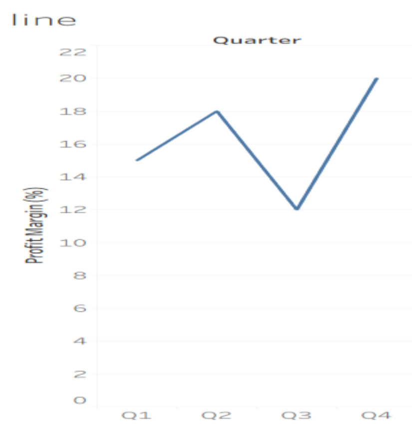
Scoring Guide

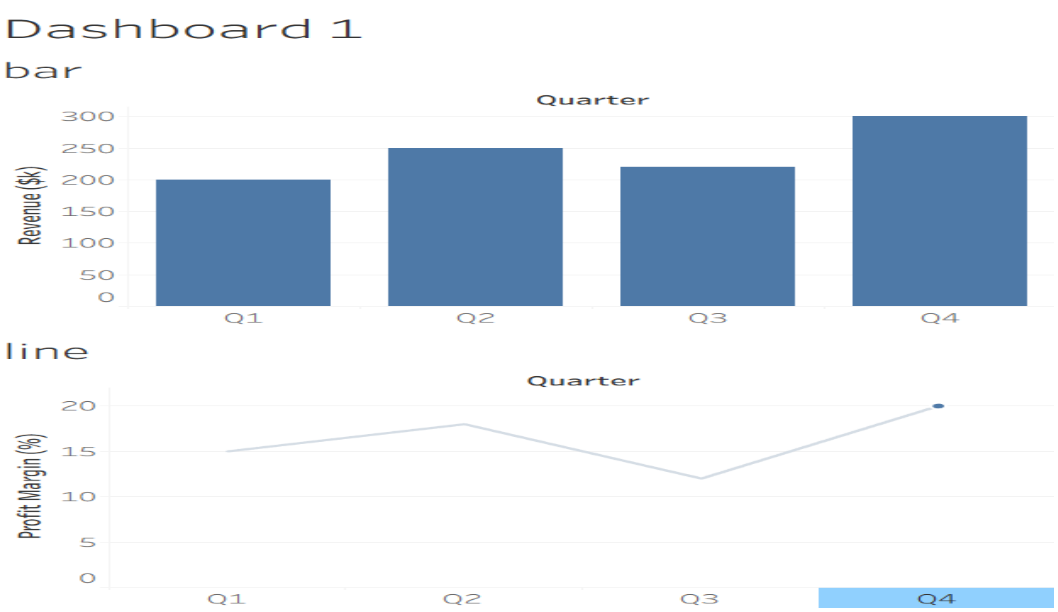
For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

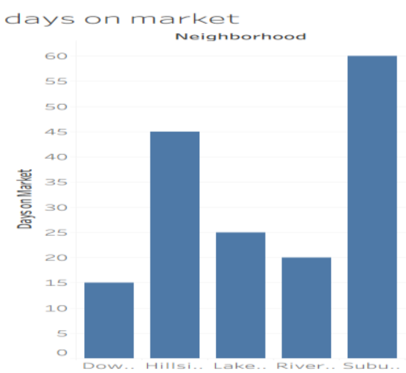
Problem 1 — Retail Quarterly Dashboard (5 Marks)
Data: Quarterly revenue and profit margin: Q1 = $200k / 15%, Q2 = $250k / 18%, Q3 = $220k / 12%, Q4 =
$300k / 20%.
Task: Build a two-chart dashboard: a bar chart of revenue and a line chart of profit margin, aligned on
the same quarter axis. Add one annotation highlighting the quarter with the highest margin, and write a
one-sentence takeaway summarizing the overall trend.

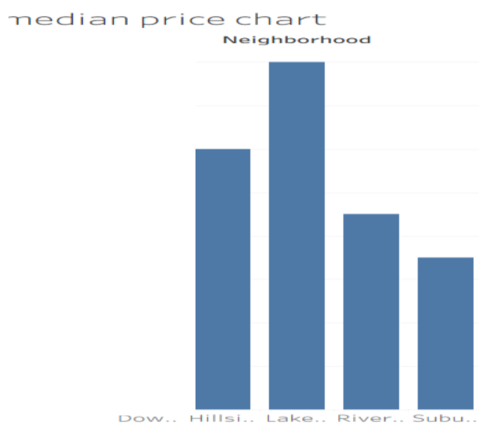


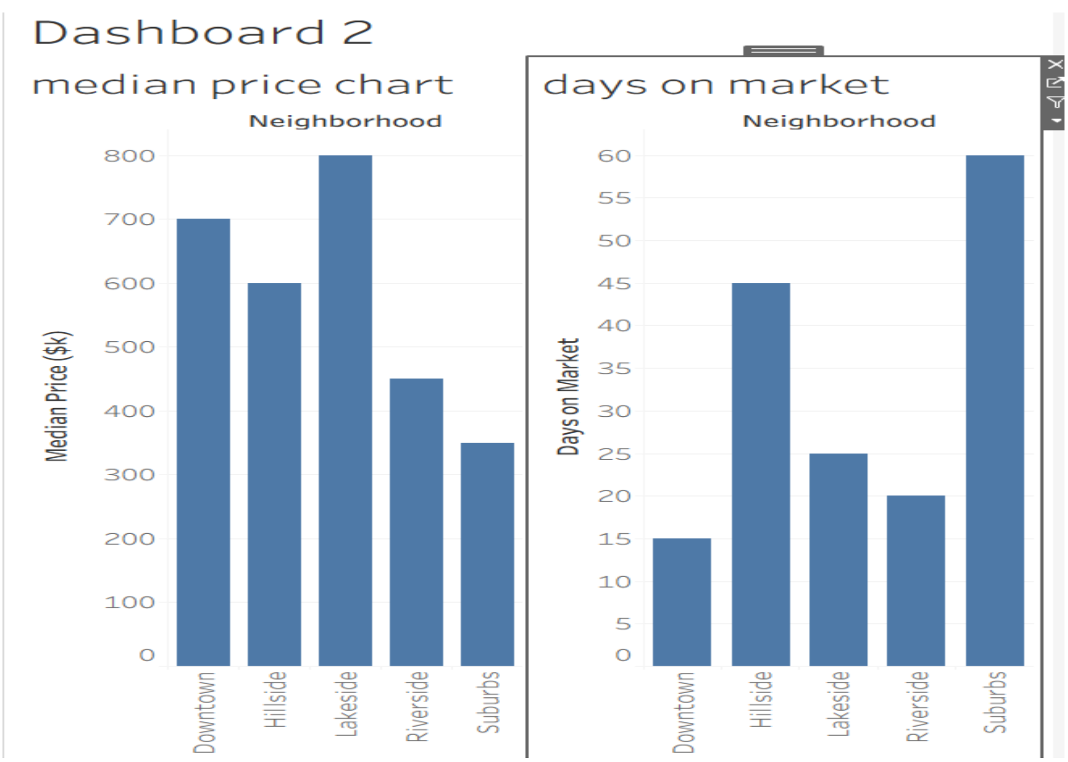




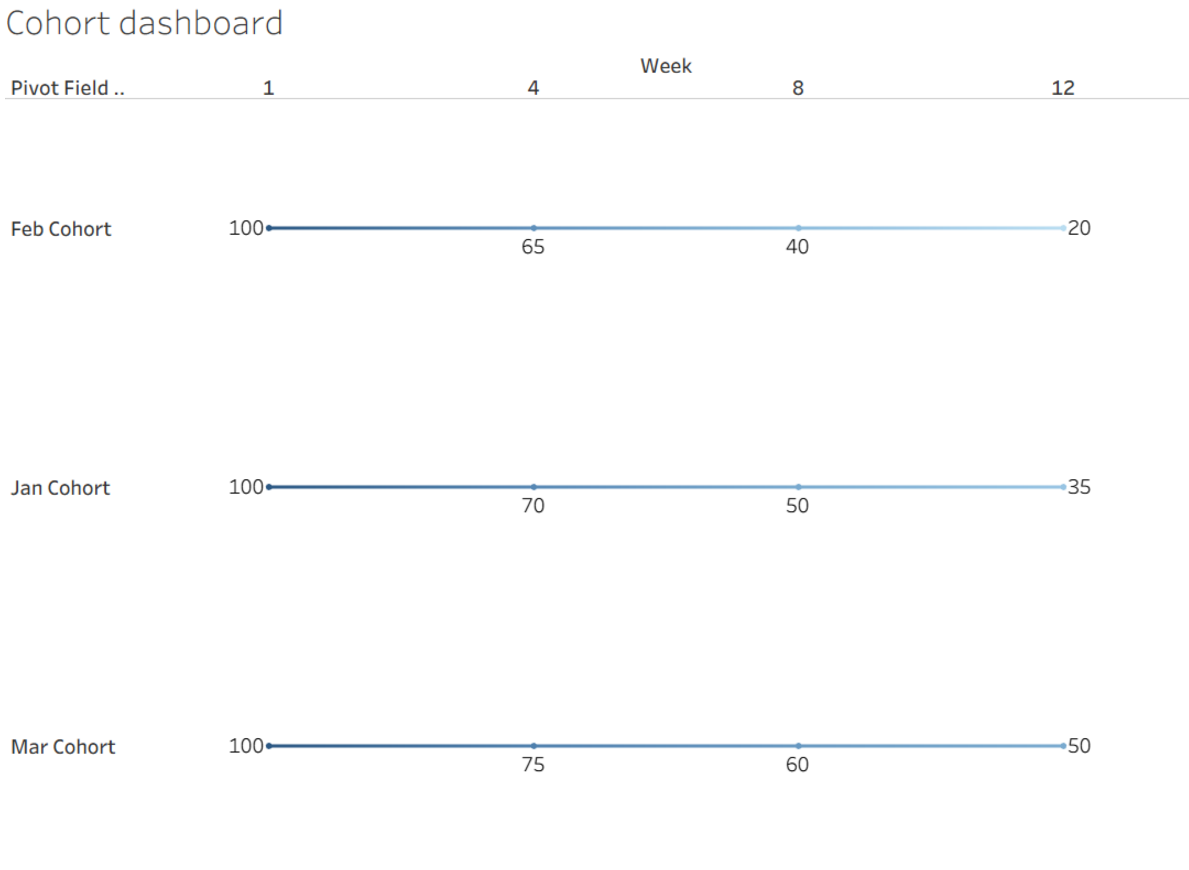
Problem 22 — Real Estate Market Dashboard (5 Marks)
Data: Median home price and days-on-market by neighborhood: Riverside = $450k / 20 days, Hillside = $600k / 45, Downtown = $700k / 15, Suburbs = $350k / 60, Lakeside = $800k / 25.
Task: Build a dashboard with a bar chart of median price and a bar chart of days-on-market for the same neighborhoods. Annotate the neighborhood with the strongest buyer demand (low days-on-market despite high price), and write a takeaway for a home buyer.



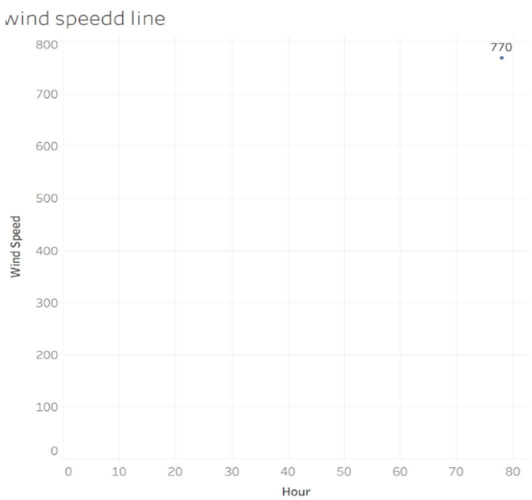


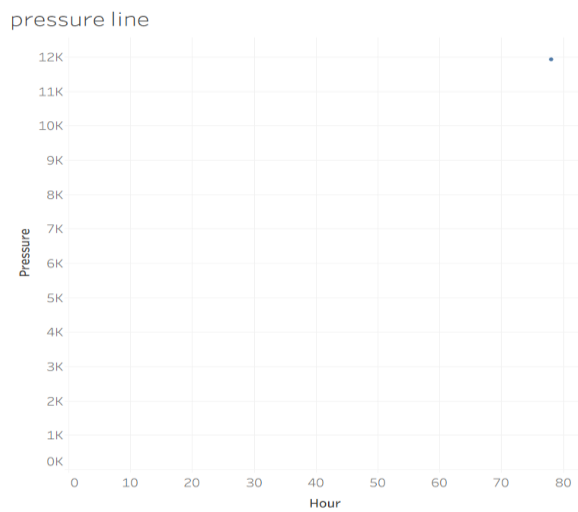


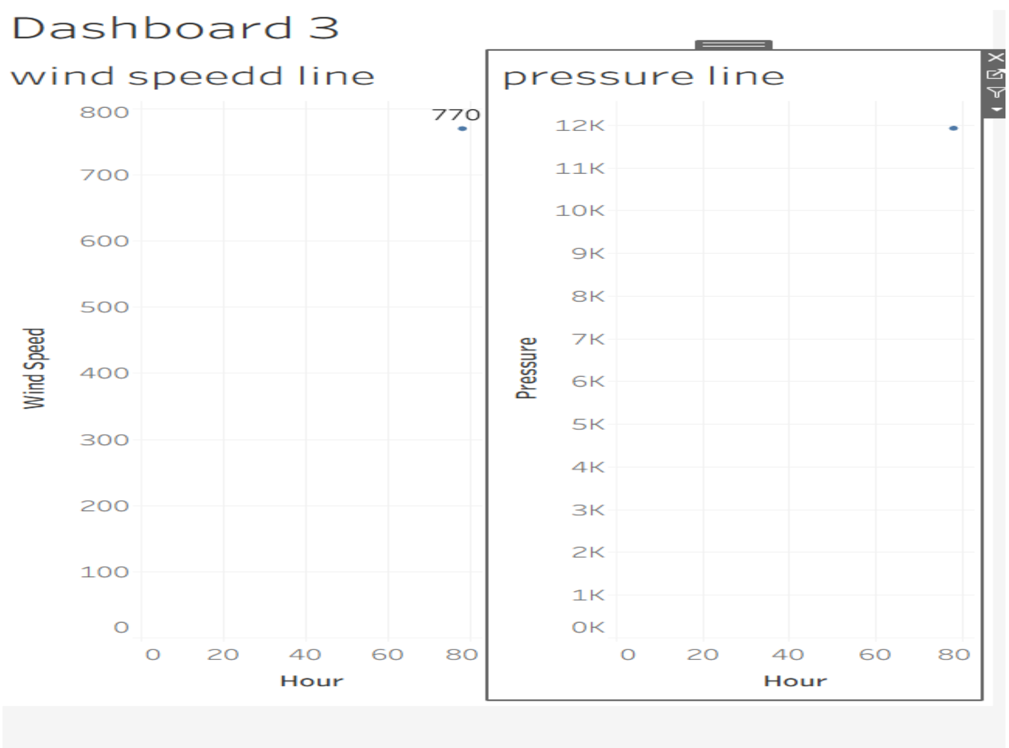
Problem 23 — Fitness App Cohort Dashboard (5 Marks)
Data: Retention rate at week 1 / 4 / 8 / 12 for three signup cohorts: Jan cohort = 100%,70%,50%,35%;
Feb cohort = 100%,65%,40%,20%; Mar cohort = 100%,75%,60%,50%.
Task: Build a dashboard with a multi-line chart showing retention curves for all three cohorts across the
four checkpoints. Annotate the cohort with the steepest drop-off, and write a narrative recommending
the best point to intervene.



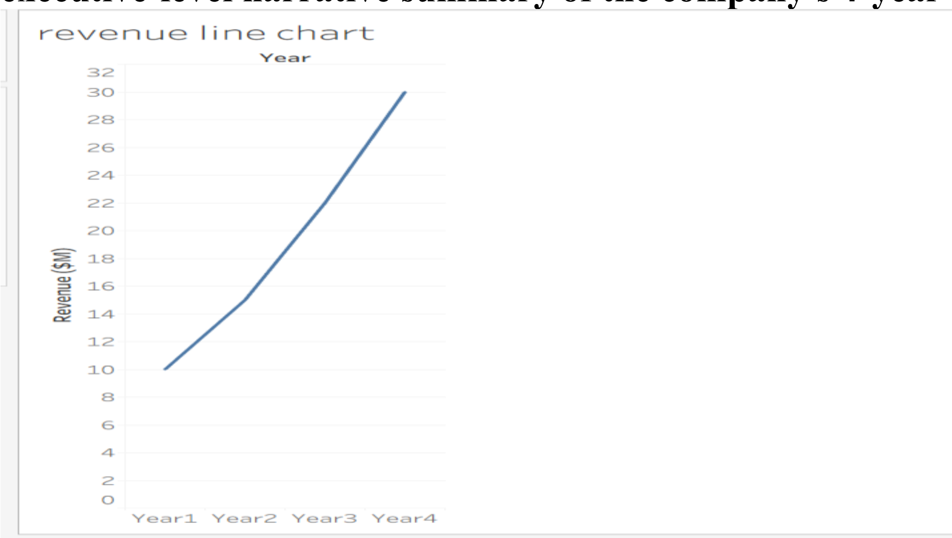
Problem 24 — Weather Storm Tracking Dashboard (5 Marks)
Data: Hourly wind speed (km/h) and barometric pressure (hPa) over 12 hours: wind =
40,45,55,70,90,110,105,85,60,45,35,30; pressure =
1005,1002,998,992,985,978,980,988,995,1000,1003,1006.
Task: Build a dashboard with a line chart of wind speed and a line chart of pressure on the same hourly
axis, stacking the two charts vertically to avoid a dual y-axis. Annotate the peak-intensity hour, and write
a narrative describing the storm's progression.







Problem 25 — Company-Wide KPI Dashboard (5 Marks)
Data: Revenue ($M), customer count (thousands), and employee count over 4 years: Revenue =
10,15,22,30; Customers = 50,80,120,180; Employees = 40,55,70,95.
Task: Build a small-multiples dashboard with three aligned mini-charts (one per metric) sharing the same
year axis. Annotate any year where growth patterns diverge across the three metrics, and write an
executive-level narrative summary of the company's 4-year trajectory.



year and employee line chart